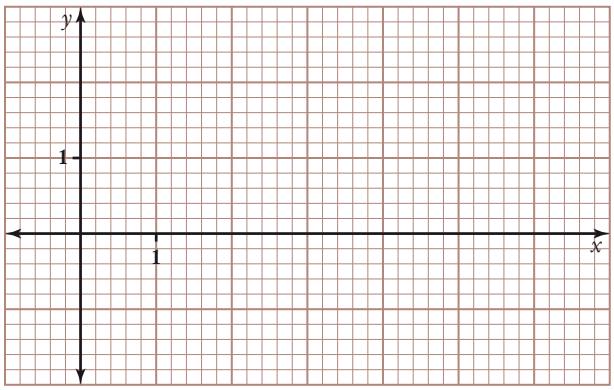
WORKSHEET

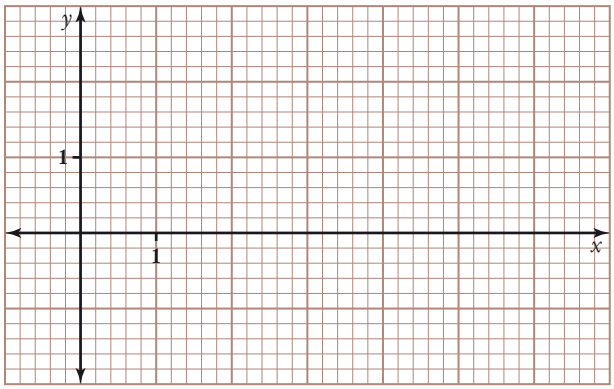
Chapter 7: Plotting log functions

1 Plot a graph to solve each of the following.

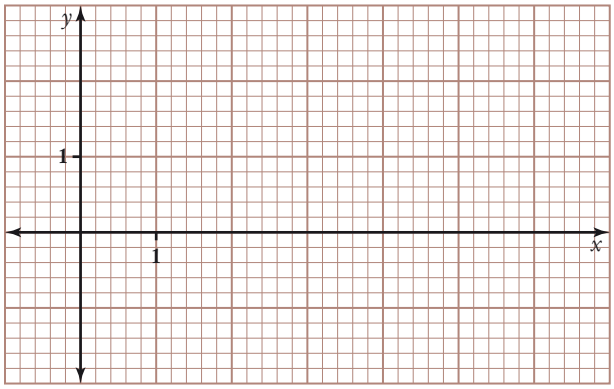
a $\log_2(x) = 2.6$



b $\log_5(x) = -0.6$

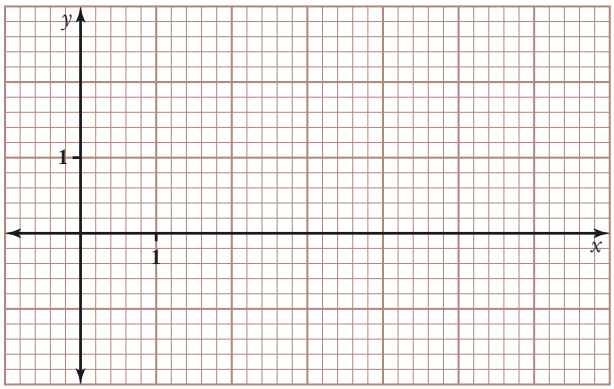


c $\log_{0.5}(x) = -1.2$



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d $\log_3(x) = 0.8$



2 Sketch the graph of each of the following, labelling important features.

a $\log_2(x) - 1$

b $\log_{0.5}(x) + 2$

c $\log_4(x) - 2$

d $\log_{0.8}(x) + 4$

e $\log_5(x) - 3$

f $\log_{0.4}(x) + 5$

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3 Sketch the graph of each of the following, labelling important features.

a $\log_2(x-1)$

b $\log_{0.5}(x+2)$

c $\log_4(x-2)$

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d $\log_{0.8}(x+4)$

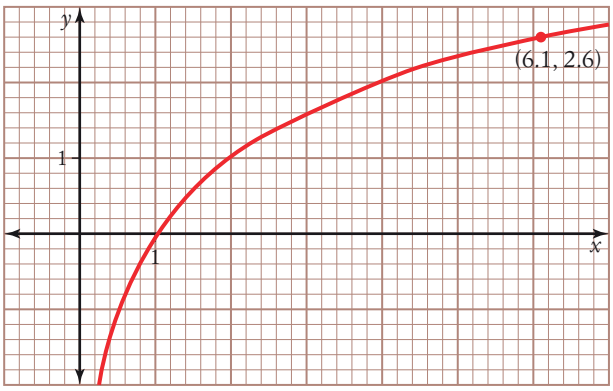
e $\log_{5}(x-3)$

f $\log_{0.4}(x+5)$

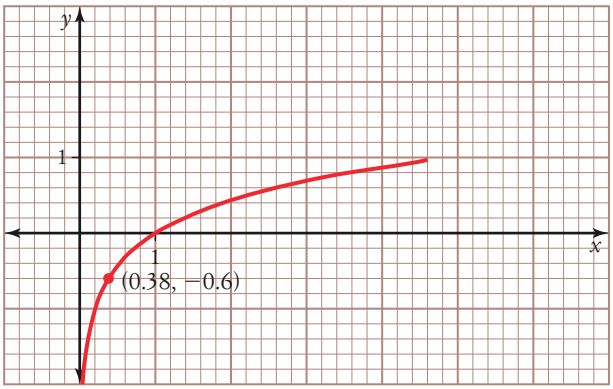
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Answers

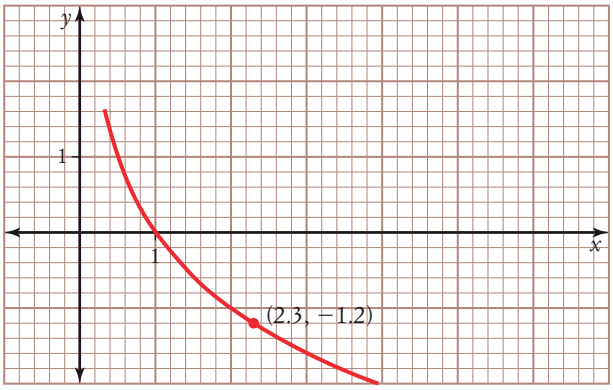
1 a $x \approx 6.1$



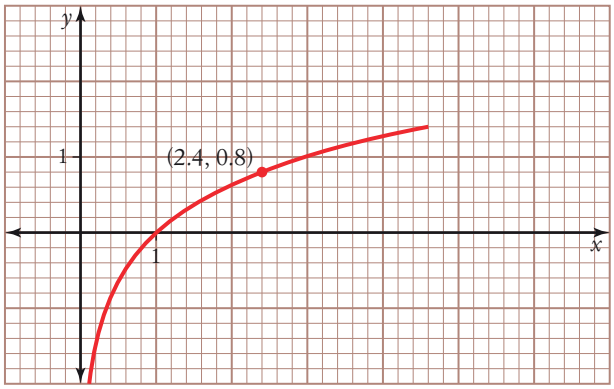
b $x \approx 0.38$



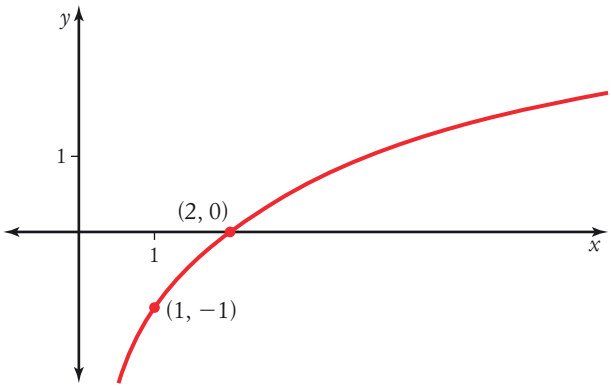
c $x \approx 2.3$



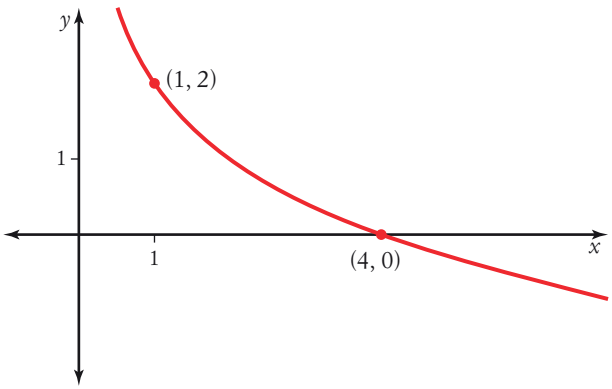
d $x \approx 2.4$



2 a Asymptote: $x = 0$

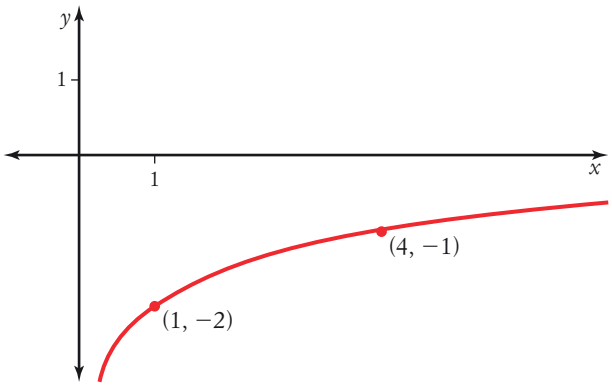


b Asymptote: $x = 0$

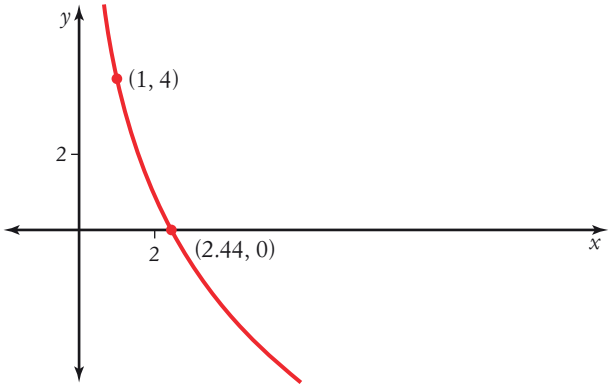


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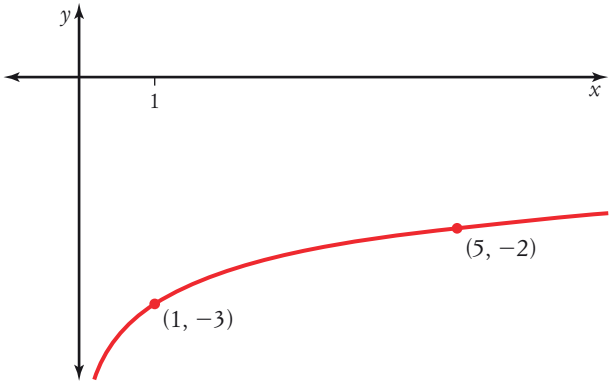
c Asymptote: $x = 0$



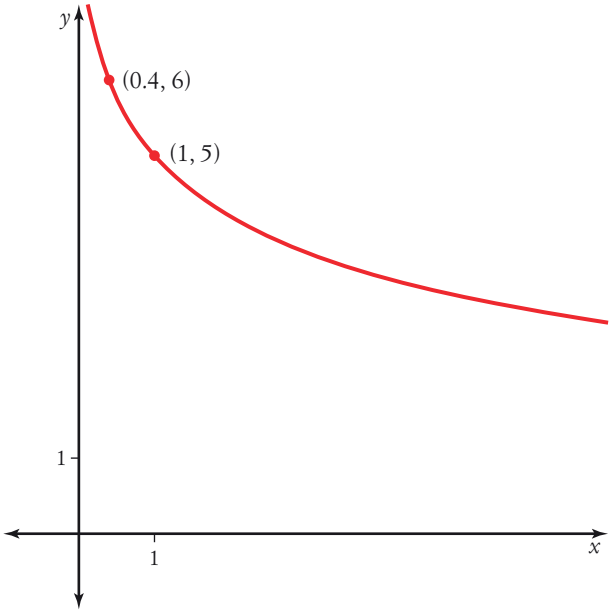
d Asymptote: $x = 0$



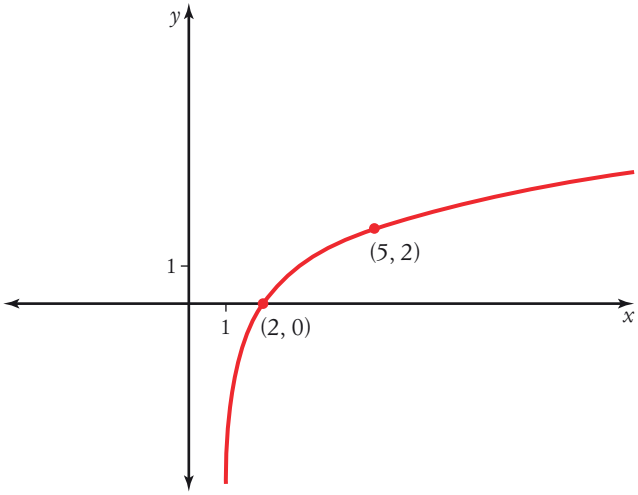
e Asymptote: $x = 0$



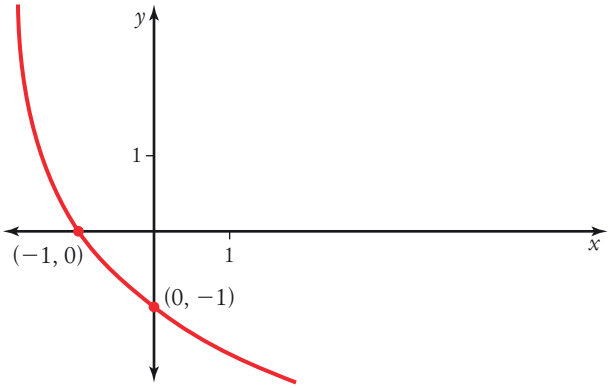
f Asymptote: $x = 0$



3 **a** Asymptote: $x = 1$

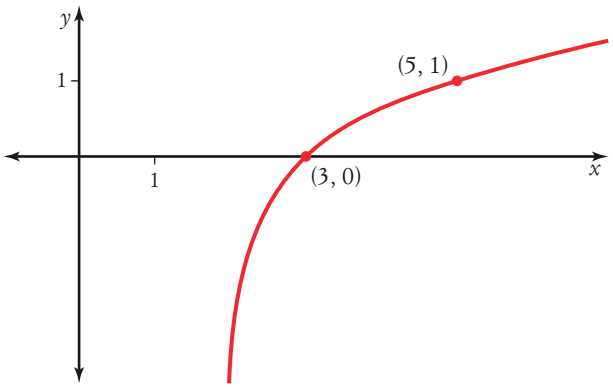


b Asymptote: $x = -2$

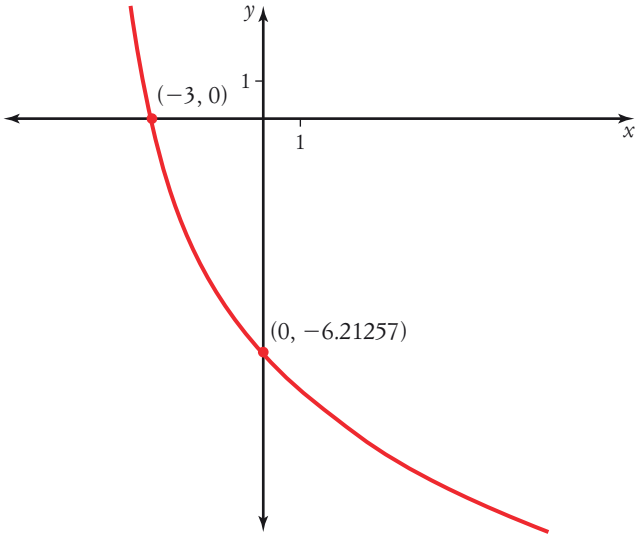


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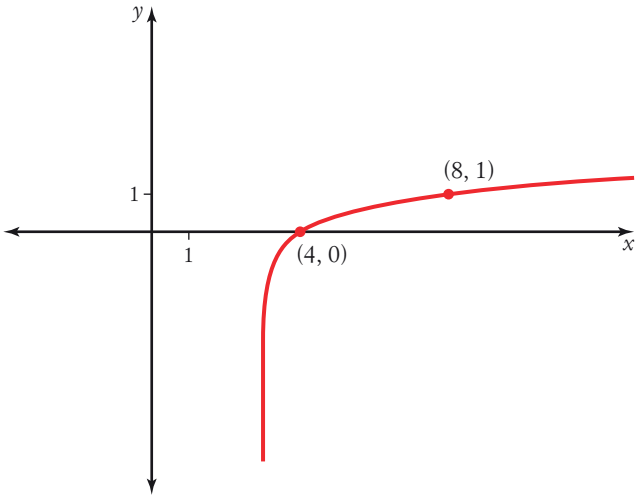
c Asymptote: $x = 2$



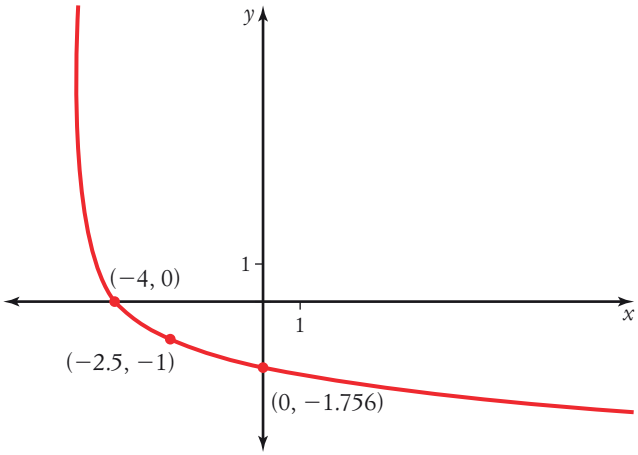
d Asymptote: $x = 3$



e Asymptote: $x = 4$



f Asymptote: $x = -5$



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